

36 Revision history

Table 206. Document revision history

Date	Revision	Changes
16-Dec-2013	1	Initial release.
08-Mar-2013	2	EMBEDDED SRAM and FLASH MEMORY: Updated Section 1.1: System architecture and Section 2.3.1: Parity check. Updated RDPRT description in Section 1.5.7: Option byte register (FLASH_OBR). Updated WRP3 at address 0x1FFF F80C in Table 8: Description of the option bytes. Updated FLASH_SR register in Section 1.6: Flash register map. Updated Table 8: Description of the option bytes. PWR: Updated Figure 1: Power supply overview (STM32F303x devices), Section 1.1: Power supplies introduction, and Section 1.1.1: Independent A/D and D/A converter supply and reference voltage. Changed AN2586 to AN4206 in Section 1.1.2: Battery backup domain. Added EWUP3, updated EWUP2 description, and added VREFINTRDYF bit in Section 1.4.2: Power control/status register (PWR_CSR). RCC: Added USART3EN, UART4EN and UART5EN in Table 1: RCC register map and reset values. Changed max. LSI clock frequency to 50 kHz in Section 1.2.5: LSI clock. Added Section 1.2.12: I2S clock (only in STM32F303xB/C and STM32F358xC). Updated Section 1.4.2: Clock configuration register (RCC_CFGR), SYSCFGRST in Section 1.4.4: APB2 peripheral reset register (RCC_APB2RSTR), SPI2RST/SPI3RST Section 1.4.5: APB1 peripheral reset register (RCC_APB1RSTR), and ADC34EN/ ADC12 EN in Section 1.4.6: AHB peripheral clock enable register (RCC_AHBENR). Replaced APB by APB2 in Section 1.4.4: APB2 peripheral reset register (RCC_APB2RSTR). Updated Section 1.4.5: APB1 peripheral reset register (RCC_APB1RSTR) and Section 1.4.2: Clock configuration register (RCC_CFGR). Updated Section 1.4.14: RCC register map.

Table 206. Document revision history (continued)

Date	Revision	Changes
08-Mar-2013	2 (continued)	GPIOs: Updated GPIOA_OSPEEDR and GPIOB_OSPEEDR reset value in Table 23: GPIO register map and reset values Replaced JTMS/SWDAT by JTMS/SWDIO in Section 9.3.1: General-purpose I/O (GPIO). Added note related to GPIOF_MODER in Section 9.4.1: GPIO port mode register (GPIOx_MODER) (x = A..F). Section 9.4.12: GPIO register map: updated GPIOA_MODER reset value, added GPIOB_MODER, GPIOA_OSPEEDR, GPIOB_OSPEEDR, GPIOB_PUPDR, update GPIOA_PUPDR reset value, updated GPIOx_LCKR register. SYSCFG: Updated Section 1.1.1: SYSCFG configuration register 1 (SYSCFG_CFGR1), Section 1.1.6: SYSCFG external interrupt configuration register 4 (SYSCFG_EXTICR4) and Section 1.1.7: SYSCFG configuration register 2 (SYSCFG_CFGR2). DMA: Updated Figure 22: DMA block diagram. Modified Figure 11.3.7: DMA request mapping. INTERRUPTS and EVENTS: Updated Table 1: STM32F302xB/C vector table. Modified address offsets for EXTI_IMR2, EXTI_EMR2, EXTI_RTSTR2, EXTI_FTSR2: 0x2C, EXTI_SWIER2, and EXTI_PR2. Updated note related to EXTI_FTSR1 in Section 13.3.4 and Section 13.3.10. Updated note in Section 1.3.7: Interrupt mask register (EXTI_IMR2). Section 1.3.13: EXTI register map ADC: Updated Section 1.1: Introduction. Updated Figure 1: ADC block diagram, Figure 3: ADC1 and ADC2 connectivity, and Figure 4: ADC3 & ADC4 connectivity. Added Section 1.3.5: Slave AHB interface. Updated caution note and note in Section 1.3.7: Single-ended and differential input channels. Modified Section 1.3.8: Calibration (ADCAL, ADCALDIF, ADC_CALFACT). Changed ADC_CALFACT_S and ADC_CALFACT_D, by CALFACT_S and CALFACT_D, respectively.

Table 206. Document revision history (continued)

Date	Revision	Changes
08-Mar-2013	2 (continued)	ADC (continued) Updated step3 of Section : <i>Converting single-ended and differential analog inputs with a single ADC.</i> Corrected the table name of Table: <i>ADC3 & ADC4 - External trigger for regular channels.</i> Removed table Minimum sampling time to be respected for fast channels and related examples in Section: <i>Channel-wise programmable sampling time (SMPR1, SMPR2)</i> and reference made to the datasheet instead. Updated Section: <i>Stopping an ongoing conversion (ADSTP, JADSTP).</i> Added note to Section: <i>Auto-injection mode.</i> Removed double buffer mode in Section: <i>Managing conversions using the DMA.</i> <i>Regular simultaneous mode with independent injected.</i> Updated Figure: <i>Flushing JSQR queue of context by setting JADSTP=1 (JQM=0). Case when JADSTP occurs outside an ongoing conversion</i> and added Section: <i>Disabling the queue.</i> Updated Section: <i>Auto-delayed conversion mode (AUTDLY)</i>, Section: DAC: Changed TIM5 into TIM15, and AIEC line9 into EXTI line9 in Table 50: External triggers. Changed TIM5 into TIM15 in the description of bits 21:19 of DAC_CR. COMP: Changed COMPx_OUT_TIM_SEL to COMPxOUTSEL in COMPx_CSR registers (x = 1 to 7) (see Section 18). Replaced COMP1SW1 by COMP1_INP_DAC in Section 15.5.1: COMP1 control and status register (COMP1_CSR). Replaced COMP2_WINDOW_MODE by COMP2WINMODE in Section 15.5.2: COMP2 control and status register (COMP2_CSR). OPAMP: Replaced AOPx by OPAMPx in Section 1: Operational amplifier (OPAMP). Updated Section 1.3: OPAMP functional description introduction and added Section 1.3.2: Clock. OPAMPx_CSR registers (Section 18.4.1 to Section 18.4.4) changed OUT-CAL, CAL_SEL, and CAL_ON to OUTCAL, CALSEL, CALON in OPAMPx_CSR registers (x = 1 to 4). Set access right to 'rw' for TSTREF. Renamed OPAMPx_EN bits into OPAMPxEN. Replaced any occurrence of VOPAMPx with VREFOPAMPx. RTC Changed power-on reset to backup domain reset in the whole section. Updated Section 24.7.15: RTC tamper and alternate function configuration register (RTC_TAFCR).

Table 206. Document revision history (continued)

Date	Revision	Changes
08-Mar-2013	2 (continued)	SPI/I2S: Remove CRC error in Section 27.6: SPI interrupts. CAN Updated Figure 317: Bit timing Advanced control timers (TIM1/8): Corrected Figure 1: Advanced-control timer block diagram. Updated Section 1.3.15: Using the break function. Updated Section 1.3.15: Using the break function General purpose timers (TIM2/3/4): Updated Section 1.4.3: TIMx slave mode control register (TIMx_SMCR). Modified UIF bit in Section 1.4.5: TIMx status register (TIMx_SR). Updated Section 1.4.6: TIMx event generation register (TIMx_EGR) and Section 1.4.9: TIMx capture/compare enable register (TIMx_CCER). General purpose timers (TIM15/16/17): Updated Figure 2: TIM16 and TIM17 block diagram. Updated Section 1.4.13: Using the break function I2C: Updated Figure 212: Setup and hold timings. Updated Table 75: Comparison of analog vs. digital filters. Corrected Figure 228: Transfer sequence flowchart for I2C master transmitter for N>255 bytes. Removed maximum values of parameter "Data hold time" and added row "Data valid time" in Table 76: I2C-SMBUS specification data setup and hold times. Added Section 25.5: I2C low-power modes. Added caution note in Section 25.4.15: Wakeup from Stop mode on address match. Moved Section 24.7: I2C debug mode to Section and renamed it Master event control using DMA. Modified definition of ARLO bit in Section 25.7.7: Interrupt and Status register (I2Cx_ISR).

Table 206. Document revision history (continued)

Date	Revision	Changes
25-Apr-2014	3	USART: Updated Section 25.5.10: LIN (local interconnection network) mode on page 663 Removed note on bit 19(RWU) in Section 25.8.8: USART interrupt and status register (USART_ISR) on page 682. Updated Table 99: USART features to remove DMA for USART5. Replaced in Bit 2 MMRQ Section 25.7.7: Request register (USART_RQR) "resets the RWU flag" by "sets the RWU flag" Added 'In Smartcard, LIN and IrDA modes, only Oversampling by 16 is supported' in Section 25.5.4: Baud rate generation Corrected and updated stop bits in Figure 198: Word length programming. TSC: Changed power-on reset value to reset value in Section 28.6.2: TSC interrupt enable register (TSC_IER). DEBUG: Updated Figure 3: JTAG TAP connections Modified scope of document from "STM32F302xx, STM32F303xx, STM32F313xx" to "STM32F303xB/xC, STM32F303x4/x6/x8, STM32F328xx and STM32F358xx". Reordered sections of the manual. Added Section 1: Overview of the manual Removed 'always read at "0" (w_r0)' in Section 1: Documentation conventions. CRC: Removed first iteration of the CRC v2 fully programmable polynomial in Section 5.2: CRC main features. EMBEDDED SRAM and FLASH MEMORY: Updated Table 2: Flash memory read protection status Updated SRAM description in Section 3.3: Embedded SRAM DAC: Added DAC2 for STM32F303x4/6/8 devices Replaced "VDDA" by "VREF+" in Section 14.3.5: DAC output voltage COMP: Added 'BRK' and 'BRK2' to the COMPxOUTSEL bit descriptions in the Section 15.5: COMP registers Updated figures in Section 15.3.1: COMP block diagram. (SYSCFG_CFGR3)

Table 206. Document revision history (continued)

Date	Revision	Changes
25-Apr-2014	3 (continued)	ADC: Updated Section 1.2: ADC main features Updated Section 1.3.6: ADC voltage regulator (ADVREGEN) Updated Section 1.3.3: Clocks OPAMP Updated Figure 3: STM32F303x6/8 and STM32F328x8 comparator and operational amplifier connections. DMA: Added Table 24: STM32F303x6/8 and STM32F328x8 DMA1 request mapping Added Table 29: STM32F303x6/8 and STM32F328x8 summary of DMA1 requests for each channel System configuration controller: Added STM32F303x4/6/8 dedicated bits in Section 1.1.1: SYSCFG configuration register 1 (SYSCFG_CFGR1) Added Section 1.1.8: SYSCFG configuration register 3 Updated Section 1.3.11: Channel selection (SQRx, JSQRx) Changed ADC1_IN18 to ADC1_IN17 in Section 1.3.31: VBAT supply monitoring Replaced DEEPPWD and ADVREGEN bits of ADCx_CR by ADVREGEN[1:0] Updated Section 1.6.2: ADC common control register (ADCx_CCR, x=12 or 34) Interrupts: Added Table 1: STM32F302xB/C vector table Removed 'or by changing the sensitivity of the edge detector.' in Section 1.3.6: Pending register (EXTI_PR1) and Section 1.3.12: Pending register (EXTI_PR2) and updated bit description of Section 1.3.5: Software interrupt event register (EXTI_SWIER1) and Section 1.3.11: Software interrupt event register (EXTI_SWIER2) Updated the programming manual reference sentence in Section 1.2.1: Main features PWR: Updated bit0 'WUF' description in Section 1.4.1: Power control register (PWR_CR) Updated the mode entry note in Table 4: Stop mode Added Figure 2: Power supply overview (STM32F3x8 devices). IRTIM: Swapped TIM16 and TIM17 in Figure 203: IR internal hardware connections with TIM16 and TIM17

Table 206. Document revision history (continued)

Date	Revision	Changes
25-Apr-2014	3 (continued)	RCC: Replaced 'PCLK' with 'SYSCLK' in the I2CxSW bits description of Section 1.4.13: Clock configuration register 3 (RCC_CFGR3) Modified note (2) under Figure 2: STM32F303xB/C and STM32F358xC clock tree. Advanced control timers (TIM1/8): Modified Section 1.3.15: Using the break function. RTC: Updated WUT input clock in Figure 207: RTC block diagram in STM32F03x, STM32F04x and STM32F05x devices Corrected bit SHPF read and clear parameters in Section 24.7.4: RTC initialization and status register (RTC_ISR) I2C: Updated tHD;DAT in Table 76: I2C-SMBUS specification data setup and hold times Replaced 50ns into tAF(min) and 260ns into tAF(max) in section I2C timings Added Access paragraph with wait state information on all registers USART: Updated Mode 2 and 3 in Section 25.5.6: Auto baud rate detection Corrected TXFRQ description in Section 25.7.7: Request register (USART_RQR) Modified Note: regarding USARTDIV in Section 26.5.4: Baud rate generation

Table 206. Document revision history (continued)

Date	Revision	Changes
18-Aug-2014	4	System and memory overview Section 3.1: System architecture System configuration controller (SYSCFG) Section 10.1.8: SYSCFG configuration register 3 (SYSCFG_CFGR3) Analog-to-digital converter (ADC) Section 13.3.18: Conversion on external trigger and trigger polarity (EXTSEL, EXTEN, JEXTSEL, JEXTEN) Figure 30: ADC1 and ADC2 connectivity Figure 31: ADC3 & ADC4 connectivity Digital to analog converter (DAC) Figure 89: DAC1 block diagram Operational amplifier Figure 104: STM32F303xB/C and STM32F358xC Comparators and operational amplifiers interconnections (part 1) Figure 105: STM32F303xB/C and STM32F358xC comparators and operational amplifiers interconnections (part 2) Comparator Section 15.5.2: COMP2 control and status register (COMP2_CSR) Digital-to-analog converter (DAC) Figure 89: DAC1 block diagram Serial peripheral interface / inter-IC sound (SPI/I2S) Table 116: STM32F303xB/C and STM32F358xC SPI implementation Universal synchronous asynchronous receiver transmitter (USART) Feature list Table 107: STM32F3xx USART features Section 27.5.2: Transmitter Section 27.7.1: Control register 1 (USARTx_CR1)
23-Jan-2015	5	Extended the applicability to STM32F303xD/E. Added Section 8: Peripheral interconnect matrix. Updated the following: Cover page Flexible memory controller (FMC) Added the chapter (applies to STM32F303xD/E only). System and memory overview Section 3.1: System architecture

Table 206. Document revision history (continued)

Date	Revision	Changes
23-Jan-2015	5 (continued)	Embedded Flash memory Section 4.1: Flash main features Section 4.2.1: Flash memory organization Reset and clock control (RCC) Section 9.2.10: Timers (TIMx) clock Section 9.4.2: Clock configuration register (RCC_CFGR) System configuration controller (SYSCFG) Section 11.1.1: SYSCFG configuration register 1 (SYSCFG_CFGR1) Section 11.1.2: SYSCFG external interrupt configuration register 1 (SYSCFG_EXTICR1) Section 11.1.7: SYSCFG register map Direct memory access controller (DMA) Table 80: STM32F303xB/C/D/E, STM32F358xC and STM32F398xE summary of DMA2 requests for each channel Interrupts and events Table 40: STM32F302xB/C/D/E vector table Analog to digital converter (ADC) Table: ADC1 (master) & 2 (slave) - External triggers for regular channels Table: ADC1 & ADC2 - External trigger for injected channels Table: ADC3 & ADC4 - External trigger for regular channels Table: ADC3 & ADC4 - External trigger for injected channels Comparator (COMP) Section 18.2: COMP main features Figure 123: STM32F303xB/C/D/E, STM32F358xC and STM32F398xE comparator 7 block diagram Table 108: Comparator input/output summary Section 18.6: COMP registers Advanced-control timers (TIM1/TIM8/TIM20) Introduced the TIM20 for STM32F303xD/E Inter-integrated circuit (I2C) interface) Table 142: STM32F3xx I2C implementation Universal synchronous asynchronous receiver transmitter (USART) Table 218: STM32L41xxx/42xxx/43xxx/44xxx/45xxx/46xxx USART features

Table 206. Document revision history (continued)

Date	Revision	Changes
25-Aug-2015	6	Serial peripheral interface / inter-IC sound (SPI/I2S) Table 167: STM32F303xB/C/D/E, STM32F358xC and STM32F398xE SPI implementation Universal serial bus full-speed device interface (USB) <i>Table: STM32F3xx USB implementation</i> Reset and clock control (RCC) Added notes about UARTSW2[1:0] and UARTSW3[1:0] availability for the STM32F3xx covered by this document in Section 9.4.13: Clock configuration register 3 (RCC_CFGR3), Updated bit 23 (V18PWRSTF) name and description in Section 9.4.10: Control/status register (RCC_CSR). Flexible static memory controller (FSMC) Renamed the section as “Static memory controller” Updated bit BUSTURN description in Section : SRAM/NOR-Flash chip-select timing registers 1..4 (FMC_BTR1..4), Updated the maximum value for memory setup time and memory hold in Table 37: Programmable NOR/PSRAM access parameters, Updated MEMSET, MEMHOLD and MEMHIZ bit descriptions of FMC_PMEM register in Section : Common memory space timing register 2..4 (FMC_PMEM2..4), Updated ATTSET, ATTHOLD and ATTHIZ bit descriptions of FMC_PATT register in Section : Attribute memory space timing registers 2..4 (FMC_PATT2..4), Updated BURSTRUN bit description of FMC_BTR1..4 register in Section : SRAM/NOR-Flash chip-select timing registers 1..4 (FMC_BTR1..4). Analog-to-digital converters (ADC) Updated the number of ADC1 and ADC2 channels for STM32F303x6/8 and STM32F328 in Table 84: ADC external channels mapping. Advanced-control timers (TIM1/TIM8/TIM20) Bit SMS description in Section 20.4.3: TIM1/TIM8/TIM20 slave mode control register (TIMx_SMCR). Basic timers (TIM6/TIM7) Updated Bit MMS description in Section 22.4.2: TIM6/TIM7 control register 2 (TIMx_CR2). General-purpose timers (TIM15/16/17) Updated IC1F[3:0] bit description in Section 23.6.6: TIM16&TIM17 capture/compare mode register 1 (TIMx_CCMR1). Universal synchronous asynchronous receiver transmitter (USART) Updated the DMA support for UART5 in Table 158: STM32F3xx USART features. Controller area network (bxCAN) Replaced tCAN with tq Universal serial bus full-speed device interface (USB) Added LPM register descriptions.

Table 206. Document revision history (continued)

Date	Revision	Changes
11-May-2016	7	Updated I2C2 section: Updated Figure 387: Setup and hold timings. Updated Section 38.4.5: I2C initialization updating and adding notes in Section : I2C timings. Updated Section 28.7.5: Timing register (I2C_TIMINGR) SCLDEL[3:0] and SDADEL[3:0] bits description. Updated Section 38.4.5: I2C initialization, Section 38.4.9: I2C master mode and Section 28.7.5: Timing register (I2C_TIMINGR) adding the sentence "The STM32CubeMX tool calculates and provides the I2C_TIMIGR content in the I2C configuration window". Updated Touch sensing controller section: Updated Section 23.3.4: Charge transfer acquisition sequence adding note about the TSC control register configuration forbidden. Updated Section 23.6.1: TSC control register (TSC_CR) adding note for CTPL[3:0] bits and PGPSC[2:0] bits. Updated USART section: Updated Section 25.5.17: Wake-up from Stop mode using USART adding paragraph "how to determine the maximum USART baudrate". Updated whole USART document replacing any occurrence of: nCTS by CTS, nRTS by RTS, SCLK by CK. Updated Section 25.8.9: USART interrupt flag clear register (USART_ICR) replacing "w" by "rc_wl". Updated Section 25.8.8: USART interrupt and status register (USART_ISR) RTOF field replacing USARTx_CR2 by USARTx_CR1. Updated Section 25.8.3: USART control register 3 (USART_CR3) 'ONEBIT' bit 11 description adding a note. Updated Section 25: Universal synchronous/asynchronous receiver transmitter (USART/UART) changing register name USARTx_regname in USART_regname. Updated RTC section: Updated WUCKSEL bits in Figure 281: RTC block diagram. Added case of RTC clocked by LSE in Section 27.3.9: Resetting the RTC. Updated Figure 281: RTC block diagram adding note. Updated Section 27.6.16: RTC tamper and alternate function configuration register (RTC_TAFCR) adding note on TAMP3 bits, and TAMP2E and TAMP2TRG in black in the register.

Table 206. Document revision history (continued)

Date	Revision	Changes
11-May-2016	7 (continued)	Updated TIMER section: Updated Section 41.3.13: One-pulse mode modifying “IC2S=01” by “CC2S=01”. Updated Section 26.3.16: Slave mode – combined reset + trigger mode adding (TIM15 only) on the title. Updated Section 133.6.8: TIM15 capture/compare mode register 1 [alternate] (TIM15_CCMR1) and Section 133.6.26: TIM15 register map replacing bit 7 ‘reserved’ by OC1CE. Updated Section 26.4.6: TIMx capture/compare mode register 1 (TIMx_CCMR1)(x = 16 to 17) and Section 26.4.23: TIM16/TIM17 register map replacing bit 7 ‘reserved’ by OC1CE. Updated Figure 24: Advanced-control timer (TIM1), Section 41: General-purpose timers (TIM2/TIM3), Section 44: Basic timers (TIM6/TIM7) and Section 26: General-purpose timers (TIM16/TIM17) PSC[15:0] bits description. Updated Section 24.4.5: TIM1 status register (TIM1_SR) and Section 119.4.65: TIM1/TIM8 register map CC5IF and CC6IF bit names. Updated FMC section: Updated Section 14.5.4: NOR flash/PSRAM controller asynchronous transactions putting ‘de-asserting the NOE signal’. Updated Section : FIFO status and interrupt register x (FMC_SRx) bit0 (IRS) and Bit2 (IFS) adding a note. Updated Section : SRAM/NOR-flash chip-select timing registers x (FMC_BTRx) adding new paragraph. Updated Section : SRAM/NOR-flash write timing registers x (FMC_BWTRx) adding new paragraph. Updated Figure 52: NAND flash/PC Card controller waveforms for common memory access replacing ‘MEMxHIZ’ by ‘MEMxHIZ+1’ and adding note 2. Updated Section 14.6.5: NAND flash prewait functionality. Updated Common memory space timing register x (FMC_PMEMx) MEMHOLD[7:0] description. Updated Attribute memory space timing registers x (FMC_PATTx) ATTHOLD[7:0] description. Updated Section 14.3: AHB interface. Updated Figure 46: Muxed write access waveforms correct NWE falling edge. Updated Embedded Flash memory: Updated Section 4.5.1: Flash access control register (FLASH_ACR) bits LATENCY[2:0] replacing SYSCLK by HCLK. Updated ADC section: Updated Section 15.3.3: Clocks note, replacing option a) by option b) and removing ‘or 10’.

Table 206. Document revision history (continued)

Date	Revision	Changes
11-May-2016	7 (continued)	Operational amplifier section (OPAMP): Updated Table 108: Connections with dedicated I/O on STM32F303xB/C/D/E, STM32F358xC and STM32F398xE. Updated DEBUG section: Updated Section 33.6.1: MCU device ID code DBGMCU_IDCODE description. Updated comparator section: Updated Figure 122: Comparator 1 and 2 block diagrams (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE) adding note 4. Updated Section 17.5.2: COMP2 control and status register (COMP2_CSR), Section 17.5.4: COMP4 control and status register (COMP4_CSR) and Section 17.5.6: COMP6 control and status register (COMP6_CSR) modifying note 3 and the bit 9 description. Updated interrupts and events section: Updated Section 13.2.6: External and internal interrupt/event line mapping line 26/28/29/31/33 and adding note for EXTI lines. Updated USB section: Updated Table 1354: STM32G4 Series USB implementation removing 'and STM32F358xC'.

Table 206. Document revision history (continued)

Date	Revision	Changes
17-Jan-2017	8	Updated comparator section: Updated Table 108: Comparator input/output summary. Updated root part numbers in: Section 18.1: Introduction. Section 18.2: COMP main features. Section 17.5.1: COMP1 control and status register (COMP1_CSR). Section 17.5.2: COMP2 control and status register (COMP2_CSR). Section 17.5.3: COMP3 control and status register (COMP3_CSR). Section 17.5.4: COMP4 control and status register (COMP4_CSR). Section 17.5.5: COMP5 control and status register (COMP5_CSR). Section 17.5.6: COMP6 control and status register (COMP6_CSR). Section 17.5.7: COMP7 control and status register (COMP7_CSR). Updated Section 17.5.2: COMP2 control and status register (COMP2_CSR) note in the bit 7 description replacing 'PA3' by 'PA7'. Added note 'depending on the product, when a timer is not available, the corresponding combination is reserved' in all the COMPxOUTSEL[3:0] bit description for x = 1...7 and COMPx_BLANKING bit description for x = 4...7. Updated Figure 122: Comparator 1 and 2 block diagrams (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE). Updated Figure 123: STM32F303xB/C/D/E, STM32F358xC and STM32F398xE comparator 7 block diagram Updated USART section: Updated Section 25.8.8: USART interrupt and status register (USART_ISR) RWU bit available independently of the wakeup from stop feature availability. Updated Section 25.8.12: USART register map RWU bit. Updated RTC section: Updated Figure 286: RTC block diagram removing 1 Hz and 512 Hz text and changing mux connections (ck_spre by ck_apre). Updated note in Section 29.4.14: Calibration clock output. Updated ADD1H and SUB1H bit descriptions in Section 29.7.3: RTC control register (RTC_CR). Updated Section : RTC backup registers and Section 29.7.20: RTC backup registers (RTC_BKPxR): RTC_BKPxR registers cannot be reset when the Flash readout protection is disabled. Updated CRC section: Added Table 140: CRC internal input/output signals in Section 20.3.2: CRC internal signals. Updated OPAMP section: Updated Figure 120: STM32F302xB/C/D/E and STM32F302x6/8 Comparators and operational amplifiers interconnections (part 1). Updated Figure 121: STM32F303xB/C/D/E and STM32F358xC cComparators and operational amplifiers interconnections (part 2).

Table 206. Document revision history (continued)

Date	Revision	Changes
08-Dec-2023	9	Updated the following sections: Section : Introduction (minor updates) Section 4.5.1: Flash access control register (FLASH_ACR) (minor updates). Section 14.3.6: Pending register (EXTI_PR1) (Reset value) Section 14.3.12: Pending register (EXTI_PR2) (Reset value) Section 4.2.3: Flash program and erase operations (Page erase, Option byte programming). Section 9.1.2: System reset (system reset introduction). Section 7.4.1: Power control register (PWR_CR) (PVD, PLS). Section 33.4.2: Flexible SWJ-DP pin assignment (Note removed). Section 7.1.2: Battery backup domain (minor update). Section 9.4.6: AHB peripheral clock enable register (RCC_AHBENR) (GPIO). Section 9.4.11: AHB peripheral reset register (RCC_AHBRSTR) (GPIO). Section 9.4.14: RCC register map (RCC_AHBENR, RCC_AHBRSTR). Section 4.3.1: Read protection (Note removed). Section 9.4.7: APB2 peripheral clock enable register (RCC_APB2ENR) (Addition of COMP on bit 0). Section 17.3.3: COMP reset and clocks (APB2, clock enable bit) Section 17.5.1: COMP1 control and status register (COMP1_CSR) (COMP1MODE[1:0]) Section 27.3.4: Real-time clock and calendar (RTCCLK) Section 9.4.3: Clock interrupt register (RCC_CIR) (Changed "xxxRDYDIE" into "xxxRDYE"). Section 6: Cyclic redundancy check calculation unit (CRC) (General updates). Section 13: Direct memory access controller (DMA) (General updates). Section 20: Advanced-control timers (TIM1/TIM8/TIM20), Section 21: General-purpose timers (TIM2/TIM3/TIM4), Section 22: Basic timers (TIM6/TIM7) and Section 23: General-purpose timers (TIM15/TIM16/TIM17) (General updates). Section 25: Independent watchdog (IWDG) (General updates). Section 28: Inter-integrated circuit interface (I2C) (General updates). Section 30: Serial peripheral interface / integrated interchip sound (SPI/I2S) (General updates). Section 29: Universal synchronous/asynchronous receiver transmitter (USART/UART) (General updates). Section 32.6: USB and USB SRAM registers (USB RAM). Minor terminology changes were applied to this document. Added the following sections: Section 35: Important security notice

Table 206. Document revision history (continued)

Date	Revision	Changes
12-Jan-2024	10	ADC: Updated Figure 54: ADC1 and ADC2 connectivity . Updated Figure 55: ADC3 & ADC4 connectivity . I2C: Whole section re-edited.

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